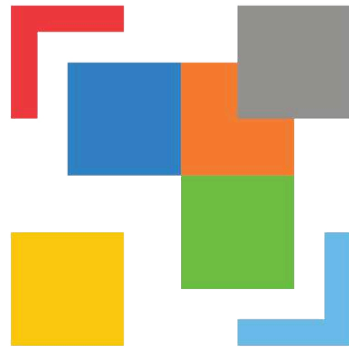




C.N.E

COMPUTER NETWORK
ENGINEERING

The general solution of $(x \cot(\frac{y}{x}) + y)dx - xdy = 0$ is

Select one:

- ☐ a. $x \sec(\frac{y}{x}) = c$
- ☐ b. $x \csc(\frac{y}{x}) = c$
- ☐ c. $x \cos(\frac{y}{x}) = c$
- ☐ d. $\cos(\frac{y}{x}) = cx$

The differential equation $y^2 y' + 3xy^3 = 3x^2 y^4$ is

Select one:

- ☐ a. Exact and Linear
- ☐ b. Separable and Nonlinear
- ☐ c. Nonlinear only
- ☐ d. Linear only

If the roots of the characteristic equation of a seventh order H.L.D.E. with constant coefficients are all repeated

$\lambda_1 = \lambda_2 = \dots = \lambda_7 = \lambda$, then the fifth solution will be of the form

Select one:

☐ a. $x^4 e^{\lambda x}$

☐ b. $x^5 e^{\lambda x}$

☐ c. $x^6 e^{\lambda x}$

☐ d. $x^7 e^{\lambda x}$

Suppose that $u = \frac{y}{x}$, then the general solution of $y' = \frac{y}{x} + \left(\frac{y}{x}\right)^3$ is

Select one:

☐ a. $\ln(x) = \frac{u^4}{4} + c$

☐ b. $\ln(x) = \frac{-1}{2u^2} + c$

☐ c. $\ln(x) = \frac{1}{u^2} + c$

☐ d. $\ln(x) = \frac{-u^2}{2} + c$

Which of the following functions is a solution to $xy' + y = 3x^2$

Select one:

- ☐ a. e^{-x}
- ☐ b. x^3
- ☐ c. x^2
- ☐ d. x

The integrating factor of $2xy' + 2xy + y = 3e^x$ is $\mu =$

Select one:

- ☐ a. $x^{-1}e^x$
- ☐ b. xe^x
- ☐ c. $\sqrt{x}e^x$
- ☐ d. $\sqrt{x}e^{2x}$

If $M(x, y)dx + N(x, y)dy = 0$ is both exact and homogeneous, then the general solution is

Select one:

- ☐ a. $xM - yN = c$
- ☐ b. $yM + xN = c$
- ☐ c. $yM - xN = c$
- ☐ d. $xM + yN = c$

If $z = y - x$, then the general solution of $y' = 1 + \frac{2(y-x)}{\ln(y-x)}$ is $x =$

Select one:

- ☐ a. $\frac{1}{4}(\ln z)^2 + c$
- ☐ b. $\frac{1}{2}(\ln z)^2 + c$
- ☐ c. $(\ln z)^2 + c$
- ☐ d. $\frac{1}{4}z^2 + c$

The general solution of $y''' - 4y'' + 5y' - 2y = 0$ is $y =$

Select one:

- ☐ a. $c_1 e^x + c_2 x e^x + c_3 e^{-2x}$
- ☐ b. $c_1 e^{-x} + c_2 x e^{-x} + c_3 e^{2x}$
- ☐ c. $c_1 x e^{-x} + c_2 x^2 e^x + c_3 e^{2x}$
- ☐ d.

$$c_1 e^x + c_2 x e^x + c_3 e^{2x}$$

The differential equation
 $xydy + (x^3y^3 - 2y^2)dx = 0$ is

Select one:

- ☐ a. Homogeneous
- ☐ b. Exact
- ☐ c. Linear
- ☐ d. Bernoulli

The general solution of $y' + \frac{5}{x}y = 2$ is $y =$

Select one:

☐ a. $\frac{1}{3x} + \frac{c}{x^5}$

☐ b. $\frac{x}{3}$

☐ c. $\frac{x}{3} + \frac{c}{x^5}$

☐ d. $\frac{x}{6} + \frac{c}{x^4}$

The general solution of the exact differential equation

$$(2ye^{2x} - y\cos(xy))dx + (e^{2x} - x\cos(xy))dy = 0$$

is

Select one:

- ☐ a. $ye^{2x} + \sin(xy) = c$
- ☐ b. $ye^{2x} - \sin(xy) = c$
- ☐ c. $ye^{2x} + \cos(xy) = c$
- ☐ d. $xe^{2y} - \sin(xy) = c$

If $y = c_1x^2 + c_2$ is the general solution to a second order D.E. Then the particular solution of the same D.E. such that $y(1) = 3, y'(1) = 2$ is $y =$

Select one:

- ☐ a. x^2
- ☐ b. $2x^2 + 2$
- ☐ c. $x^2 + 1$
- ☐ d. $x^2 + 2$

If the roots of the characteristic equation of a H.L.D.E. with constant coefficients are $1, -2$, then the D.E. is

Select one:

☐ a. $y'' + y' + 2y = 0$

☐ b. $y'' - y' + 2y = 0$

☐ c. $y'' - y' - 2y = 0$

☐ d. $y'' + y' - 2y = 0$